

HIGH SPEED, HIGH POWER USB C PD HUB

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Submitted for ECSE 398 in partial fulfillment for the degree of Bachelor of Science of Electrical Engineering

Case Western Reserve University

Fall 2024

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Three handwritten signatures in black ink, likely belonging to the students Zach Brown, Toby Cowles, and Carter Graefnitz, positioned below the academic integrity statement.

Executive Summary

This document details the design and development of a high-speed, high-power USB-C Power Delivery (PD) hub. The primary objective of this project was to create a USB hub capable of delivering high power (up to 100W per port) and high-speed data (10Gb/s) simultaneously, overcoming limitations found in existing commercial solutions.

Design and Implementation:

The project involved selecting appropriate Integrated Circuits (ICs) to meet speed and power requirements, designing a block diagram, and capturing a schematic for the hub. The design was then laid out onto a multilayer PCB, which was reviewed and validated by stakeholders.

Benefits:

The proposed hub offers a highly integrated solution that leverages USB-C and USB 3.2 Gen 2 capabilities without unnecessary features, reducing cost and complexity. It provides a viable alternative to existing solutions which either lack power delivery or data transfer capabilities.

Technical Constraints:

Challenges included limited IC selection for high-speed USB hubs and the complexity of routing high-frequency data traces. The design adhered to USB-IF and IPC standards to ensure signal integrity and manufacturability.

Results and Verification

The project successfully produced a PCB design that integrated all target functionalities and met design rule checks. Some issues with component availability and clearance were noted, but the design was manufactured successfully with minor adjustments.

Budget and Project Management

The project was managed using a Gantt chart and communication tools, with a budget of \$636.01 for five assembled units. The project remained within our initial estimated budget, demonstrating effective cost management.

Conclusion

The USB-C PD hub project achieved its primary objectives, providing a functional and efficient design. Further work is needed to complete assembly and functional validation. The project represents a significant step forward in developing versatile and high-performance USB hubs.

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Problem Statement:

There is not a commercial USB hub available that is capable of supplying high power USB C PD and high speed data (10Gb/s+) on the same hub. Therefore we propose the design and manufacture of such a hub.

Approach:

Our recommended approach is as follows:

First we will research the relevant USB-IF specifications and select ICs capable of implementing a hub that will support USB 3.2 gen 2, as well as 100W USB PD.

We will design a block diagram including all of these ICs in order to better understand the system level approach. Next we will capture a schematic of the hub that implements the recommended ICs, and review it with the stakeholders as well as other electrical engineers.

Afterwards we will lay out the schematic onto a multilayer PCB, and have the layout reviewed by stakeholders and other electrical engineers.

Finally we will have the board fabricated and assembled by a Contract Manufacturer (CM), and shipped to CWRU for bring-up and validation.

Benefits:

The benefit of our solution compared to commercial solutions is that our solution takes full advantage of the capabilities offered by USB C and USB 3.2 Gen 2 for data and power without any unnecessary features that would increase cost and complexity with no benefit for many applications. The specific commercial competition can be broadly categorized as falling into the following categories, each with a specific downfall compared to our product.

Competition:

- USB Hubs: Can't carry power beyond the standard 5V 1A of USB 3
- USB C PD Multiport Chargers: Can't carry data
- Thunderbolt/USB Docks: Complicated, expensive, limited number of PD ports compared to other ports

Additionally our design approach offers the benefit of a highly integrated and polished solution capable of supporting this niche functionality, and we have IP for boards that can be used to order more boards as needed with ease, and can be tweaked or expanded if required in the future.

Purpose, Goals & Objectives:

Requirements:

Requirement type:	Min Spec:	Notes
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Speed	10GB/s (USB 3.2 Gen 2)	
Power output	100w per port USB-C PD at 5, 9, 15, and 20V.	
Power input	12V/24V DC, 400w	We wanted to ensure our hub could operate across a range of input voltages to allow for multiple power supply battery configurations
Number of ports	4 DFPs, 1 UFP	
Size	<160mm x 160mm>	This is a general guideline and note a hard requirement

Table 1. Design requirements.

Success Criteria

Design Criteria:

The design will be considered successfully if it:

Criterion 1. A completed PCB design is produced

Criterion 1.1. The PCB integrates all target functionality

Criterion 1.2. The PCB design can be exported as a gerber design for assembly

Criterion 2. The board completes a DRC with no serious violations

Criterion 2.1. All trace widths are within manufacturer capabilities

Criterion 2.2. All high speed traces meet impedance targets ¹

Criterion 2.3. All components do not violate pad to pad or courtyard clearances

Criterion 2.4. No traces or components violate board edge clearances

Criterion 3. The design passes the manufacturer DFM

Criterion 3.1. No issues that prevent manufacturability are present in the PCB routing

¹ High speed differential pairs must be perfectly impedance matched to 90 ohms; the single-ended configuration channel signals for USB-C were also impedance matched to 50 ohms.

Criterion 3.2. All components can be assembled successfully by our CM

Functional Criteria:

The project will be considered to be successful if the finished module:

Criterion 1. Establishes successful USB data and PD links to a variety of devices.

Criterion 2. Meets the relevant test assertions laid out in the following test specifications in addition to any other documents we may find useful by USB-IF:

Criterion 2.1. [USB Type-C and Power Delivery Source Power Requirements Test Specification](#)

Criterion 2.2. [USB 3.2 Link Layer Test Specification](#)

Criterion 2.3. [USB Type-C Functional Test Specification](#)

Criterion 3. Stays within a safe temperature range

Criterion 3.1. All ICs and switching components should have a surface temperature of less than 80C under full load

Criterion 3.2. All user accessible surfaces should have a surface temperature of under 55C

Criterion 4. Provides a viable alternative to commercial USB PD + data solutions

Technical Constraints

Early on in the project we identified some technical constraints that would drive some elements of the design.

TC 1. Limited IC selection. There are few 10gb/s USB hub ICs on the market and none faster. Making custom ICs/working with non-public ICs is out of scope of this project so we will be constrained by what is available.

TC 1.1. We will choose ICs that are capable of meeting our speed target of $\geq 10\text{gb/s}$

TC 1.2. We will choose ICs that are easily available through our contract CM

TC 2. Routing traces and manufacturing boards for high frequency data gets more difficult the higher the data speed, which could limit the CMs we can work with as well as impose limits on the design tools required.

TC 2.1. Stackup will follow Microchip Inc. and USB IF recommendations for weave to ensure good signal integrity. This includes selecting a thin dielectric between layer 1 and ground, selecting at least 1 oz./ft^2 copper weight, and selecting a substrate with low tangent losses and a uniform dielectric constant.

TC 2.2. USB differential pairs will be impedance controlled to $90\text{ ohm} \pm 7\text{ ohms}$

TC 2.3. USB diff pairs will be referenced to a ground plane immediately below or above them.

TC 2.4. USB CC lines will be impedance controlled to $50\text{ ohm} \pm 5\text{ ohms}$

TC 3. The ICs we choose should be able to work together within the USB C, USB 3.2, and USB PD specifications/standards.

TC 3.1. Only one IC may communicate with the downstream device

TC 3.2. USB 3.2 data should be MUXed prior to the USB C port to allow reversibility based on the direction detected by the CC lines

TC 3.3. The full USB PD voltage range (5-20V) should be supported based on negotiation over the CC lines

Understanding these technical constraints informed our design approach, we knew that choosing the right ICs was the foundation the rest of the project would depend on. We also knew that in order to achieve success we needed to be able to design and manufacture a board that could carry

the required data speeds. We also knew that in order to choose the right ICs we needed to understand the standards.

Standards

FCC:

Using the definitions provided in 47 CFR part 15 subsection 3 of the FCC's rules, we can classify our device as both a "digital device" and an "unintentional radiator." The frequency of the digital clock signals utilized in our device operate above 9kHz, requiring it to be classified as a "digital device" under paragraph (k). Additionally, because our device transmits radio frequency signals via copper traces without the intention "to emit RF energy by radiation or induction," it falls under paragraph (z), classifying it as an "unintentional radiator" [1].

- [1] *FCC Radio Devices Rule on Definitions*, 47 C.F.R § 15.3, 2024. [Online] Available: <https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-A/section-15.3>. Accessed November 20, 2024.

Although this project was inspired by a missing commercial solution in mind, our intent for the design and manufacturing is largely developmental and we do not intend to sell or market the device. We do not expect this design to be widely distributed and have only ordered 5 devices for manufacturing. This device was designed as part of an academic exercise with the expectation that it will only be used within a controlled laboratory environment for further academic research. For these reasons, this device falls under subsection 23 [2], allowing it to be classified as a home-built device, exempting it from the majority of regulations applied to the classifications in subsection 3 paragraphs (k) and (z) [1] since we have employed the best engineering practices to our knowledge as required by subsection 23 [2].

- [2] *FCC Radio Devices Rule on Homebuilt Devices*, 47 C.F.R § 15.23, 2024. [Online] Available: <https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-A/section-15.23>. Accessed November 20, 2024.

Home-built devices, however, must still comply with the conditions of 47 CFR 15.5 [3]. Since our device classifies as an unintentional radiator, subsection 5 requires any user of our device to cease operations upon notification of harmful interference due to the device's radio frequency emissions [3].

- [3] *FCC Radio Devices Rule on General Conditions of Operation*, 47 C.F.R § 15.5, 2024. [Online] Available: <https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-A/section-15.5>. Accessed November 20, 2024.

USB-IF:

The PHY implementation in the PCB layout was implemented following recommendations provided in several USB-IF standards.

- [4] *Universal Serial Bus 3.2 Revision 1.1 Specification*, USB-IF, June 2022. [Online]. Available: <https://usb.org/document-library/usb-32-revision-11-june-2022>. Accessed November 20, 2024.

This revision of the USB 3.2 Specification covers architecture, protocols, and data transfer models, focusing on the enhanced SuperSpeed bus architecture, error handling, power management, and backward compatibility with USB 2.0. The specification outlines detailed requirements for device manufacturers to ensure compliance and interoperability within the USB ecosystem, including mechanical, physical, and electrical layer standards, as well as packet transmission and error recovery mechanisms. Particularly relevant for our project is the way SuperSpeedPlus hubs exhibit distinct behaviors compared to previous types of USB hubs. SuperSpeedPlus hubs, operating at Gen 1x2 speeds or beyond, such as the gen2 hub used in this project, utilize a store-and-forward model, enabling the reception of multiple packets before retransmitting. This design supports more efficient packet scheduling across multiple devices. Additionally, both hub types manage transitions from low-power states and preserve packet order during data transmission.

- [5] *USB Power Delivery Specification Revision 3.2*, USB-IF, Oct. 2023. [Online]. Available: <https://usb.org/document-library/usb-power-delivery>. Accessed November 20, 2024.

This revision of the USB Power Delivery Specification provides detailed descriptions of power modes and the contract negotiation process in the USB Power Delivery Specification. Describes the system operates in different power modes, such as Standard Power Range (SPR) and Extended Power Range (EPR), each requiring specific contract negotiation. For SPR the fixed voltage levels include 5V, 9V, 15V, and 20V. The associated current levels vary, with maximum currents of up to 3A for most voltage levels. In specific cases, a 5A cable can allow for higher current delivery. The programmable power supply (PPS) mode for SPR mode supports voltage levels ranging from 5V to 21V with similar current levels depending on the power profile (15W, 27W, 45W, 60W, 100W). In an explicit contract negotiation, the source advertises its power capabilities, and the sink evaluates and requests the appropriate power levels, followed by the source acknowledging the request and adjusting the power accordingly. This contract negotiation is essential for ensuring optimal power delivery. Furthermore, implicit contracts are used temporarily, such as during power role swaps, but they are immediately followed by explicit contract renegotiation

- [6] *USB Type-C Cable and Connector Specification Release 2.3*, USB-IF, Oct. 2023. [Online]. Available: <https://usb.org/document-library/usb-type-cr-cable-and-connector-specification-release-23>. Accessed November 20, 2024.

This USB Type-C specification explains in detail the role and capabilities of both the USB Type-C cable and receptacle. Our interest will be in understanding the configuration channel (CC) and its role in determining plug orientation, sink/source power roles, and UFP/DFP data roles. Particularly important to our project, the CC is where contract negotiations take place for USB PD implementation and the determination of Vconn. This standard also points out the differences between USB Type-C cables, noting that cables can have differing capabilities relating to the

supported version of USB, cable length, current ratings, and supported PD ranges. To meet the USB 3.2 Gen 2 and 100W PD capabilities of our hub, a CC3G2-5 Type-C cable or higher must be used.

IPC:

[7] *IPC-2221: Generic Standard on Printed Board Design (IPC-2221B), Revision B*. IPC, 2012.

IPC-2221 is the industry standard for the design and layout of printed circuit boards. It is a massive document that details everything from the selection of materials and foils for a PCB design to the differences in topology of the solder on a HASL leaded versus lead free pad. 2221 is applicable to our project mostly because we are developing a sophisticated, high speed and high power PCB. 2221 will assist us in ensuring that our board is able to meet the demanding requirements of our project while still being meeting manufacturability constraints and remaining affordable.

[8] *IPC-6012: Qualification and Performance Specification for Rigid Printed Boards (IPC-6012C), Revision C*. IPC, 2010.

IPC-6012 goes over the criteria for the design process, materials, testing and finally acceptance of PCBs to ensure reliability and quality. 6012 covers electrical integrity, durability, environmental stresses, and general performance. We selected 6012 specifically because of the demanding environment that our final hardware will face, and its need to be able to withstand high vibration environments and specific shock load cases.

[9] *IPC-2141: Controlled Impedance Circuit Boards and High Speed Logic Design (IPC-2141A), Revision A*. IPC, 2004.

IPC-2141 goes over selecting trace geometry to meet the demands of high speed and controlled impedance boards. It focuses on microstrip and stripline waveguides, as well as outlining guidelines for developing functional and manufacturable high speed PCBAs. We selected IPC-2141 as our design is very high speed, and a majority of the traces on our board will be serving as transmission lines that need to be impedance and length matched.

Background and Context:

Project Overview

The design process for our hub started with our conceptual design phase where we researched different options for components. Using this initial set of components, we drew out a system block diagram.

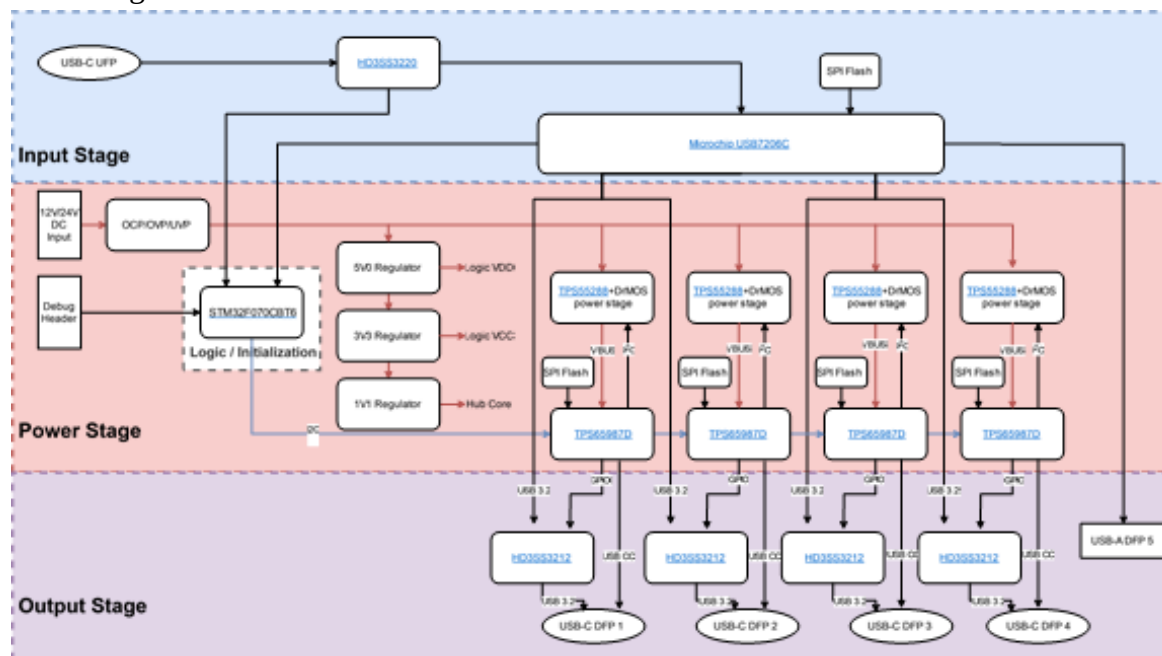


Figure 1. Architectural block diagram.

We ultimately selected the Microchip USB7206C as our main hub IC for several reasons. First, Microchip was one of the few companies offering publicly available USB 3.2 Gen 2 hub ICs. Second, this IC was well-documented and featured five USB 3.2 Gen 2 downstream facing ports (DFPs) and one USB 2.0 DFP. Four of the USB 3.2 DFPs were connected to Texas Instruments (TI) HD3SS3212 MUX ICs. These MUXes were chosen because they supported USB 3.2 Gen 2 speeds and did not rely on CC lines for direction detection. Instead, direction detection could be set via an external GPIO. To control the MUXes' directionality and handle power delivery (PD) negotiation on the CC lines, we used the TI TPS65987D. This highly versatile IC can directly control the MUXes through GPIO and manage the TI TPS55288 digital buck/boost converter over I²C. As a result, our hub is capable of functioning as a fully integrated hardware device compliant with the USB specification. However, to add flexibility, we also included an STM32F070CBT6 microcontroller (MCU). This MCU enables configuration and bootstrapping of the other ICs as needed.

Power Architecture

One of the most critical components of the hub project is the PD subsystem. This subsystem is responsible for listening for a power delivery (PD) request over the CC pins, negotiating a PD contract with the downstream device, telling our digital buck-boost converters to go to this set voltage, and finally closing the load switches to connect the converter to the downstream port. There are a few tricky pieces to this task. First, the regulators must be able to provide anywhere from 5 volts to 21 volts at up to 5 amps. With an input voltage of 12 volts, we need a solution capable of either “bucking” the voltage down from 12V to 5V, or “boosting” the voltage up to 21V. The solution we ended up with is a buck-boost controller from TI, the TPS55288. This controller is a 4-switch buck-boost converter with an I²C interface that allows our PD port controller, the TPS65987D, to pass the voltage requested by the downstream port to the regulator. The TPS65987D is responsible for ensuring that the output voltage and current of the regulator are within the ranges specified by the contract, and will open its load switches in the case that an overcurrent or incorrect output voltage is detected.

We also simulated the hardware design of the buck-boost converter using TI’s Webench power designer. The plots below show some key performance metrics about the regulator design.

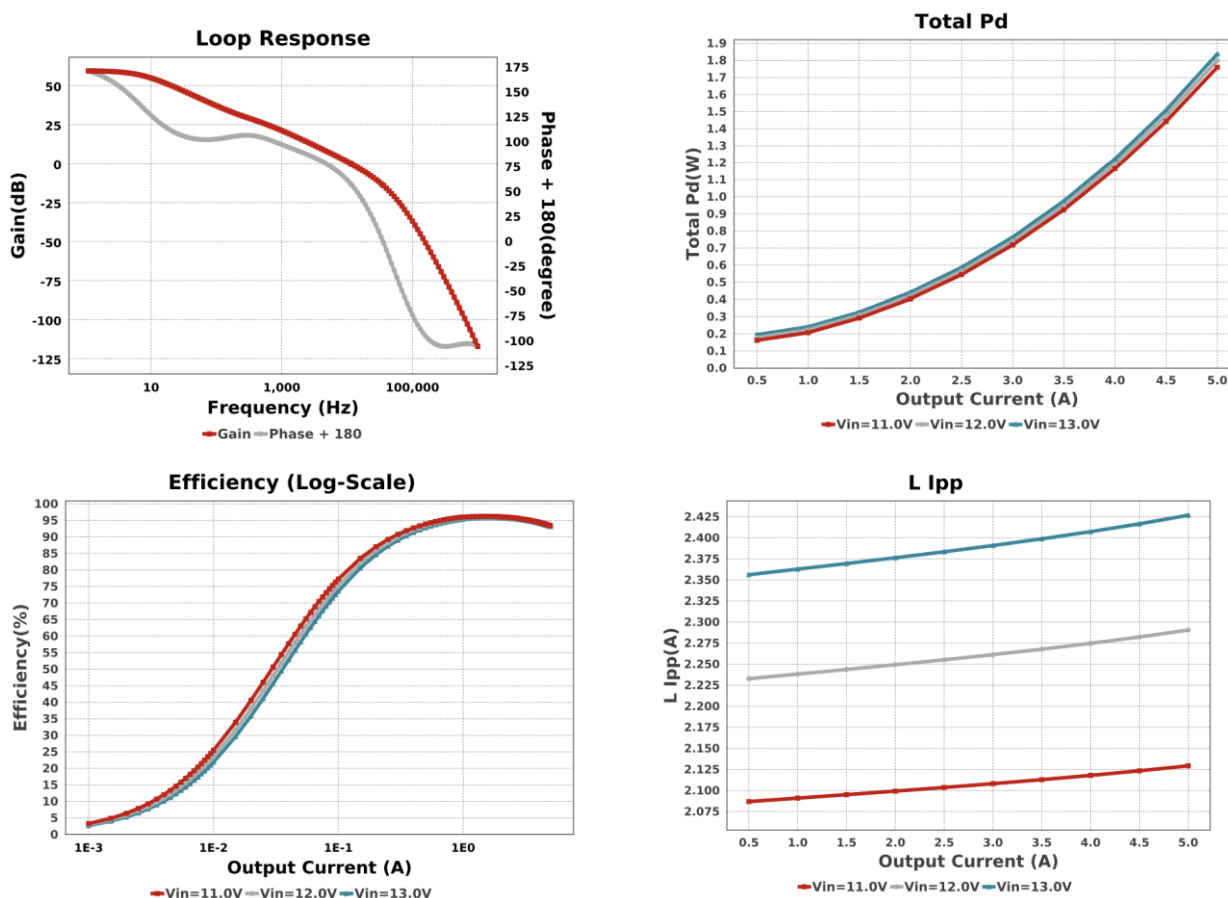


Figure 2. Closed loop response, total power dissipation, efficiency, and inductor current of the regulator.

Design Methodology, Approach, & Implementation:

Design Tools

We evaluated multiple different EDA (Electronic design automation) tools to complete our PCB design, we were initially planning on using KiCAD, as it is free, powerful, and fairly intuitive, however we ultimately chose to complete this project in Altium designer due to its superior routing capabilities for impedance controlled traces as well as it's built in collaborative version control system (Altium 365), it's support for multichannel designs and our ability to access free student licenses.

Schematic Capture

We completed the schematic capture using Altium's multi channel design feature, this allowed us to design major blocks of our design on separate sheets which we could then turn into sheet symbols to integrate together. Thumbnails of the schematic sheets are attached here to provide an overview of the design structure. Full sheets are available in the appendix.

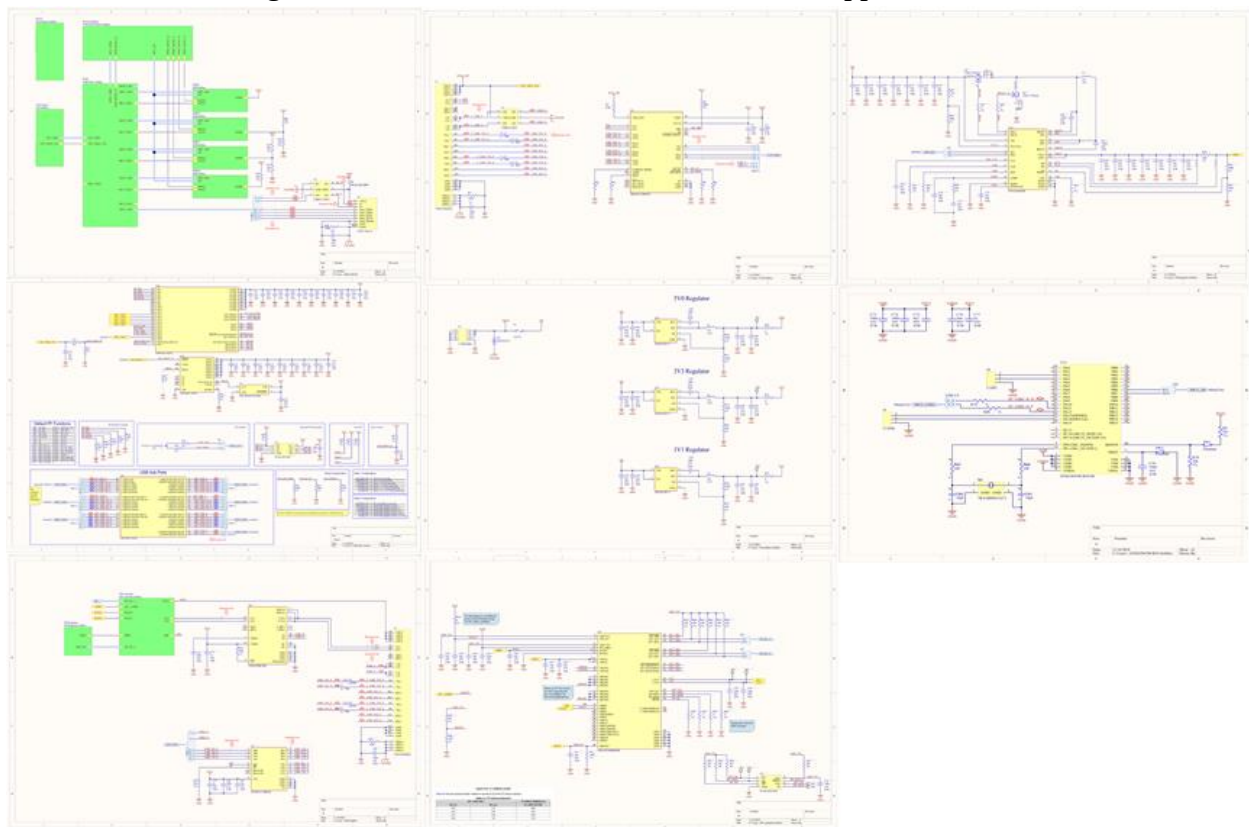


Figure 3. All schematic sheets.

PCB

After schematic capture we were able to import the netlist into a PCB document. We completed the initial layout of each 'room' or block of the multichannel design which could then be

Figure 4. PCB Layout

Assembly/Fabrication

We designed our PCB with fabrication in mind, from the very start we researched and chose a CM [contract manufacturer] with the capability to fabricate and assemble our boards, ultimately settling on JLC PCB, a well established player with a good range of capabilities, low prices and quick turnaround times. We made several design decisions to make the manufacturing process as painless as possible. We chose parts that would be available in our CMs parts library and added JLC part numbers to our BoM sheet, additionally we chose our layer stack based on JLCs options settling on the JLC0616H-7628 stack up.

×

Layers: 6 Thickness: 1.6 Outer Copper Weight: 1 Inner Copper Weight: 0.5

No requirement	JLC06161H-3313	JLC06161H-7628	JLC06161H-1080
JLC06161H-2116A	JLC06161H-3313A	JLC06161H-1080A	JLC06161H-1080B
JLC06161H-7628A			

Figure 5. Layer stackup options

We identified this stack up prior to starting our PCB design based on the speed requirements of our design. Our digital traces will see an analog bandwidth of anywhere from 15-25 GHz, which

means the design is extremely sensitive to the characteristics of the transmission lines that make up the data traces on the board. Any small inconsistencies in the dielectric constant of the substrate that the copper foil is glued to can cause impedance changes that will negatively affect the signal integrity of the USB3 traces. We selected a 7628 weave for the fiberglass prepreg and core layers of our PCB as it is the most dense and uniform glass weave that JLCPCB, our CM, offers. A 1080 weave would have many small transitions between glass fiber and epoxy that would create changes in the Dk of the transmission line, and thus its impedance, which would then create signal integrity problems due to reflections, radiation, and lower signal strength. We then had the option of the spacing between layers. Changing this spacing influences the required trace width to hit a specific impedance, with a thicker substrate increasing the impedance of a fixed-width trace. We selected the thinnest spacing between our outside layers and their ground planes to best control dissipation in the weave as well as use trace widths that suited our design constraints. Below is the final stackup entered into Altium, which then uses a 2D EM field solver to calculate the required trace widths to achieve our target impedances.

#	Name	Material	Type	Weight	Thickness	Dk	Df
	Top Overlay		Overlay				
	Top Solder	Solder Resist	Solder Mask		0.01016mm	3.8	
1	SIG 1		Signal	1oz	0.03556mm		
	Prepreg 1	PP-006	Prepreg		0.21041mm	4.4	0.02
2	GND 1	CF-004	Signal	1/2oz	0.0175mm		
	Core 1	PP-006	Core		0.4mm	4.6	0.02
3	SIG 2	CF-004	Signal	1/2oz	0.0175mm		
	Prepreg 2	FR-4	Dielectric		0.20279mm	4.4	
4	PWR	CF-004	Signal	1/2oz	0.0175mm		
	Core 2	PP-006	Core		0.4mm	4.6	0.02
5	GND 2	CF-004	Signal	1/2oz	0.0175mm		
	Prepreg 3	PP-006	Prepreg		0.21041mm	4.4	0.02
6	SIG 3		Signal	1oz	0.03556mm		
	Bottom Solder	Solder Resist	Solder Mask		0.01016mm	3.8	
	Bottom Overlay		Overlay				

Figure 6. In depth layer stackup

Design Changes:

Over the course of this project we made several changes.

Our initial project plan involved using Microchip USB7252C hubs, as they offered internal support for USB-C, eliminating the need for external MUXes. However, this approach had significant drawbacks compared to the final design we adopted. First, the design required two hub ICs, which was not ideal. Second, both the PD controller ICs and the hub ICs needed access to the CC lines, as the hubs relied on these lines for direction detection. This overlap created complications in the design. We addressed these issues by switching to external MUX ICs, which allowed the PD controller IC to manage directionality. This adjustment enabled us to consolidate the design, using a single hub to support all the necessary ports.

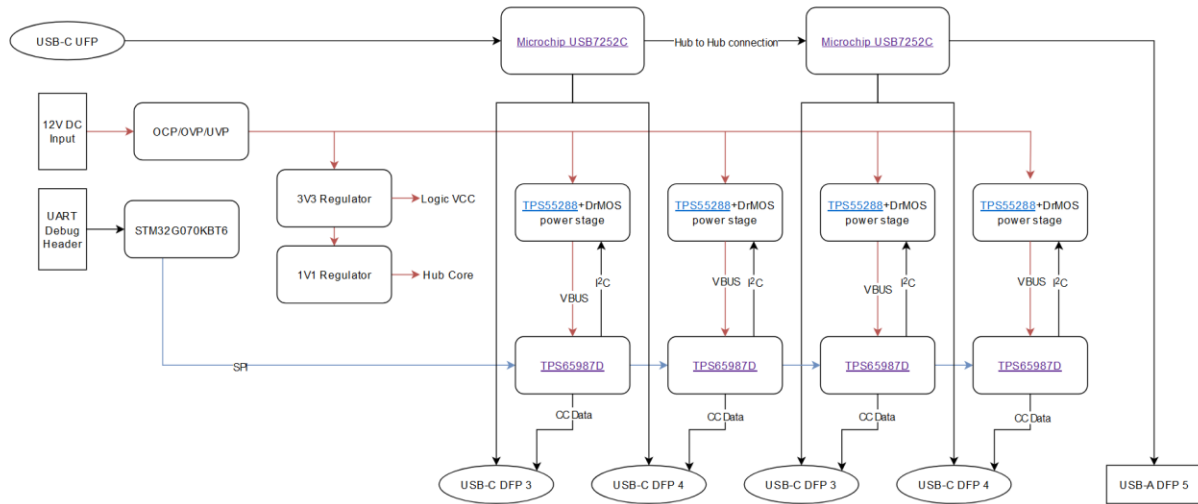


Figure 7. Initial Block Diagram

This block diagram went through several additional revisions before we settled on the final version, but these were mostly minor variations.

Verification & Results:

Design Verification:

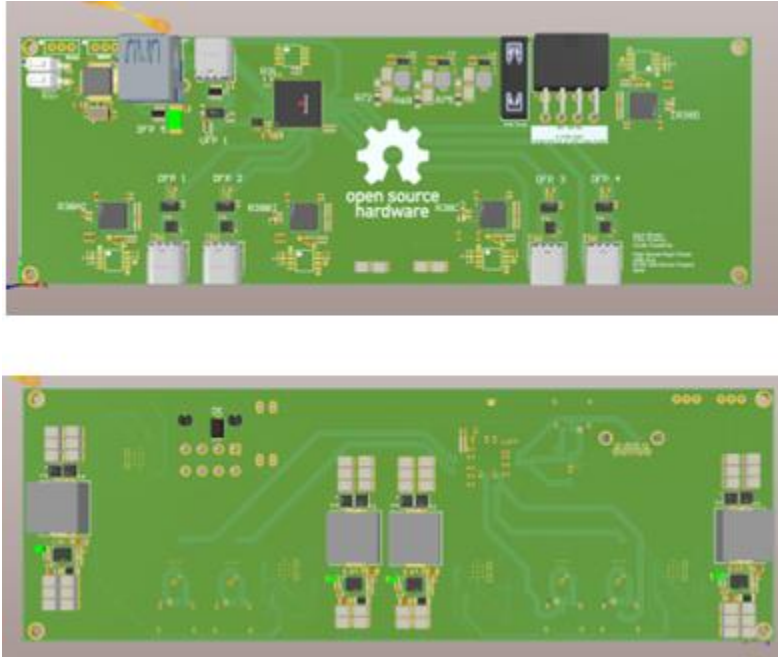


Figure 8. 3D rendering of PCB layout

Results and Statements of Success:

Criterion 1. A completed PCB design is produced

Criterion 1.1. The PCB integrates all target functionality:

This success criteria was met, as we were able to select and integrate ICs for all functionality

Criterion 1.2. The PCB design can be exported as a gerber design for assembly:

This criteria was achieved, all assembly files including GerberX2, BoM and CPL files were exported successfully.

Criterion 2. The board completes a DRC with no serious violations

Criterion 2.1. All trace widths are within manufacturer capabilities:

This criteria was successfully met. Traces rules were set according to manufacturer spec and no violations were identified after DRC was ran

- Criterion 2.2. *All high speed traces meet impedance targets:* ²
This criteria was conditionally met. All 90 ohm differential pair traces were within their design rule tolerance, however the 50 ohm single ended CC lines raised 40 DRC violations due to small sections on the top layers where the impedance was 54.8 ohms, however due to the necessity of these narrower traces to interface with the pad on the footprint for the IC, the small length of these sections, and the low speed of the CC lines this was determined to be within tolerance and the violations were waived.
- Criterion 2.3. *All components do not violate pad to pad or courtyard clearances:*
This success criteria was conditionally achieved . Altium did not raise any DRC violations however a violating component was later identified during the manufacturing process.
- Criterion 2.4. *No traces or components violate board edge clearances:*
This success criteria was achieved successfully.

Criterion 3. The design passes the manufacturer DFM

- Criterion 3.1. *No issues that prevent manufacturability are present in the PCB routing:*
This was achieved successfully, our CM JLC successfully manufactured our PCB without issue.
- Criterion 3.2. *All components can be assembled successfully by our CM:*
This was not fully met. Some components were not available in the CM part library due to stock and time issues, additionally we ran into clearance issues not caught by Altium on U6 and C32 pictured below.

² High speed differential pairs must be perfectly impedance matched to 90 ohms; the single-ended configuration channel signals for USB-C were also impedance matched to 50 ohms.

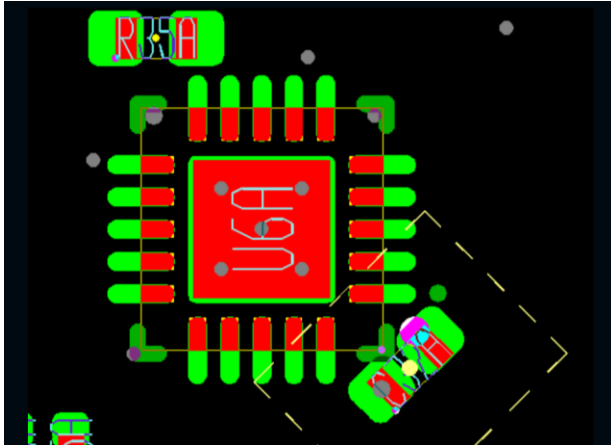


Figure 9. The component clearance violations between U6A and C32.

Additional Analysis:

We fully completed the design including schematic capture, PCB layout, and routing and this design was able to be exported for manufacturing, meeting success criteria one. Prior to our final export of the design we spent some time doing review and verification. Over the course of our review and verification, DRC identified 95 possible violations. After a manual review of these violations it was determined 40 of these violations are nets in the CC class not meeting the defined rules for their target impedance due to space constraints, and should be able to be safely ignored. 25 of these are net antenna violations caused by TI provided footprints, and can also be ignored. The remaining 20 are small annular rings in the Amphenol provided footprints on the USB C connectors and can also be safely waived. For these reasons we consider criterion 2 to be conditionally met. After working with our contract manufacturer we were able to produce a series of boards successfully, though due to stock issues combined with time constraints and a clearance violation not all components were assembled by our contract manufacturer. For this reason we consider criterion 3 to be partially met.

Remaining work to do:

We have remaining work to do in completing the assembly, bring up, and functional validation of the boards. This is reflected in the functionality success criteria which we were not able to verify this semester.

Project Management

Our project management this semester started with clear communications between team members. We wrote a project proposal and a project requirement doc to make clear what our goals and deliverables were. We then broke the project into phases which could be entered into Open Project to create a Gantt chart.

We also used less formal methods of project management including a Discord group to help facilitate communication between team members.

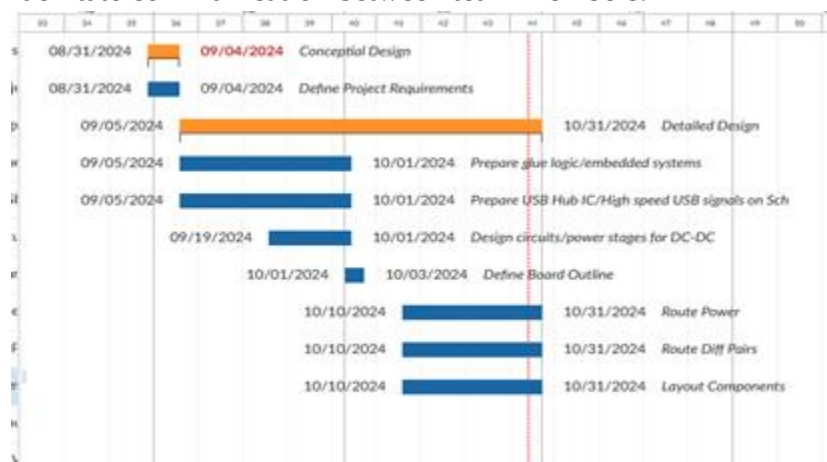


Figure 10. Gantt Chart

Budget:

We made early estimations as a ballpark for the budget seen below. This was a cost estimate for 3 units.

Part	Estimate	Note
BOM	125	This is per unit. Main IC's are \$62 per unit. Assume an addition \$63 per unit for passives, DrMOS power stages, and any support ICs,
PCBA	300	\$143 for PCB (5 units) \$50 for shipping \$100 for PCBA (3 units) NOTE: Due to minimum order quantity PCB and PCBA estimate is for multiple units
Total	$300 + 125 \times 3 = \text{\$675 total (3 units)}$	\\$225 per unit (for 3 unit)

Table 2. Initial Project Budget

After completing the design and placing the order the final merchandise total came to \$636.01 for 5 assembled units. This includes the \$78.29 for PCBs, and \$520.30 for PCBA, which includes both the BoM and assembly costs. However this does not include the USB C connectors, or fuses, which would add an additional approximately \$50, however we are still in line or better then our original budget goals given that we are under \$700 for 5 units.

Relevant Courses

Course	Description
ENGR 210	Basic circuit and network theory, use of instrumentation
ECSE 245	More advanced power circuitry,
ECSE 309	Covers high speed/high frequency signals and the concept of impedance matching
ECSE 344	More advanced power circuitry
ECSE 371	Applied circuit design
ECSE 303	Firmware development

Table 3. Courses upon which our team drew on to complete this project.

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Figure A1. Main Schematic

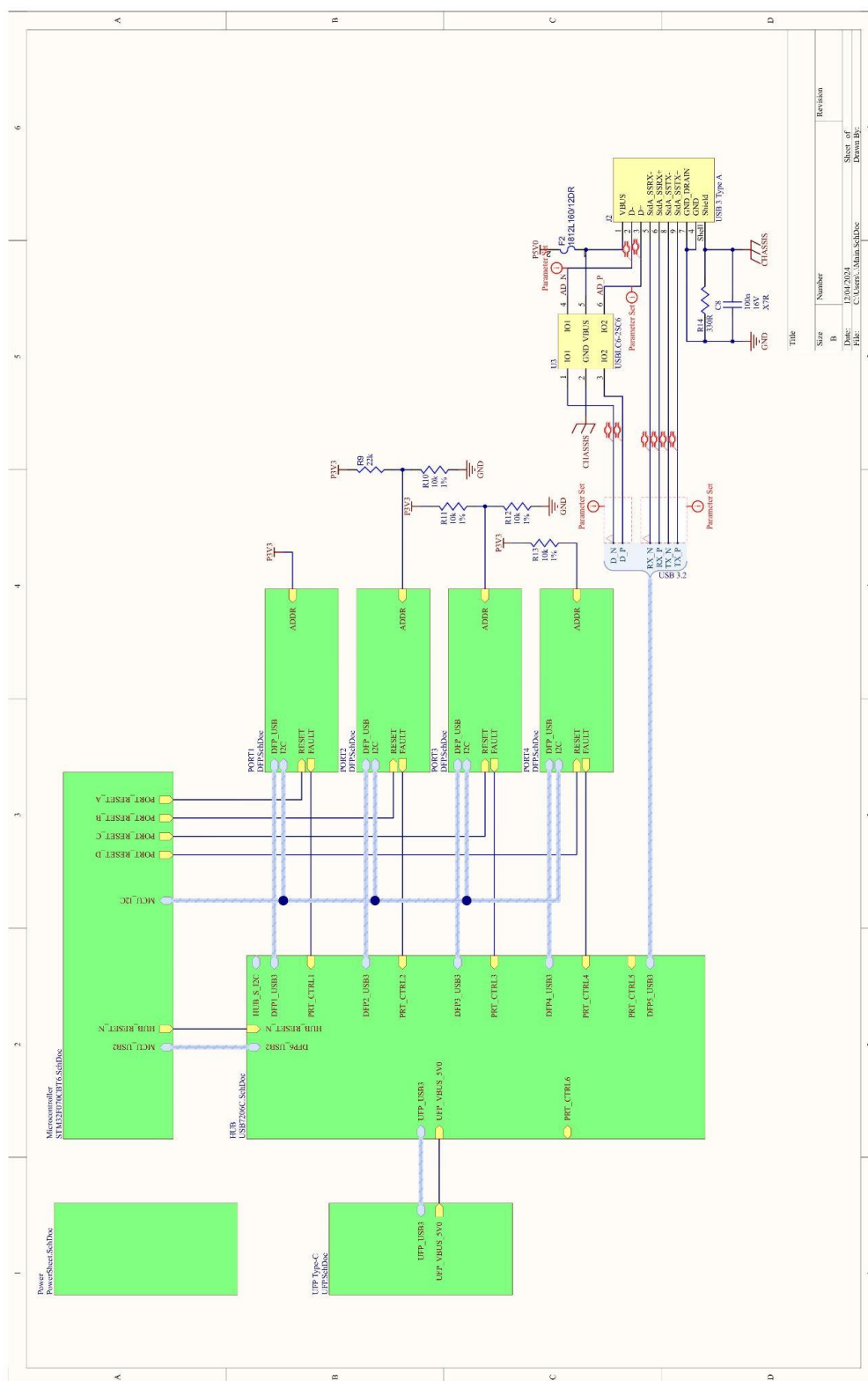


Figure A2. Power Schematic

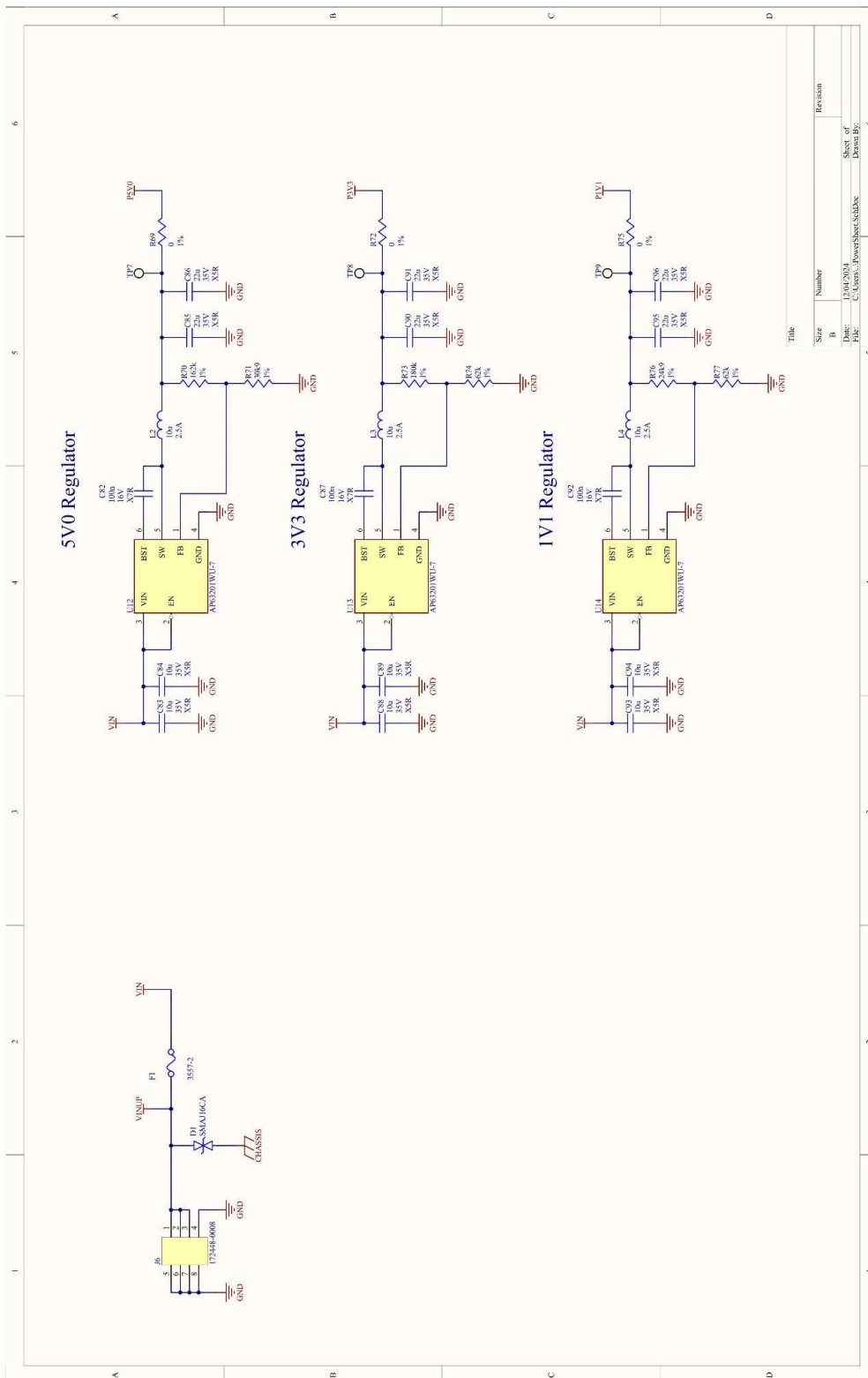


Figure A3. UFP Schematic

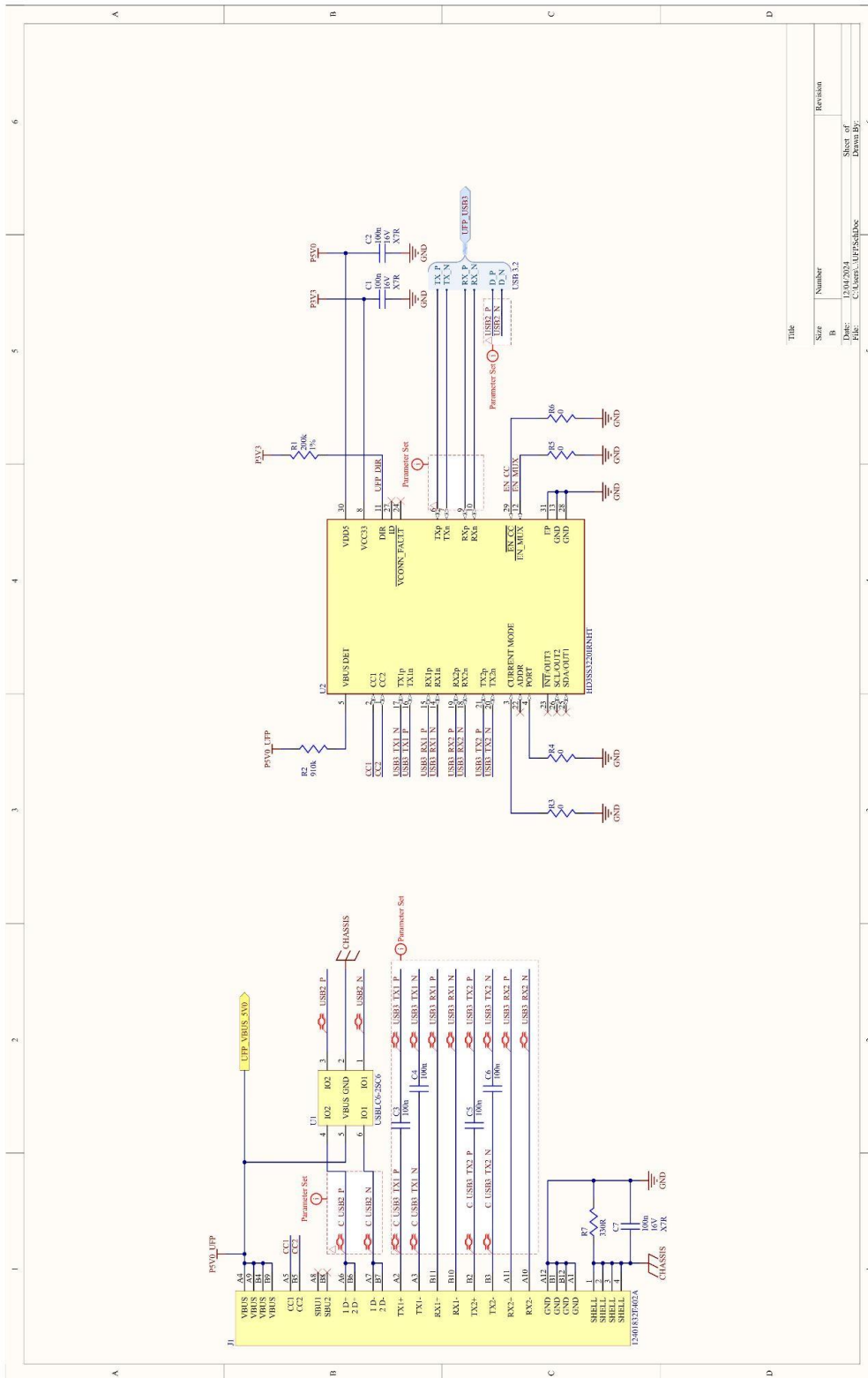


Figure A4. USB Hub Schematic

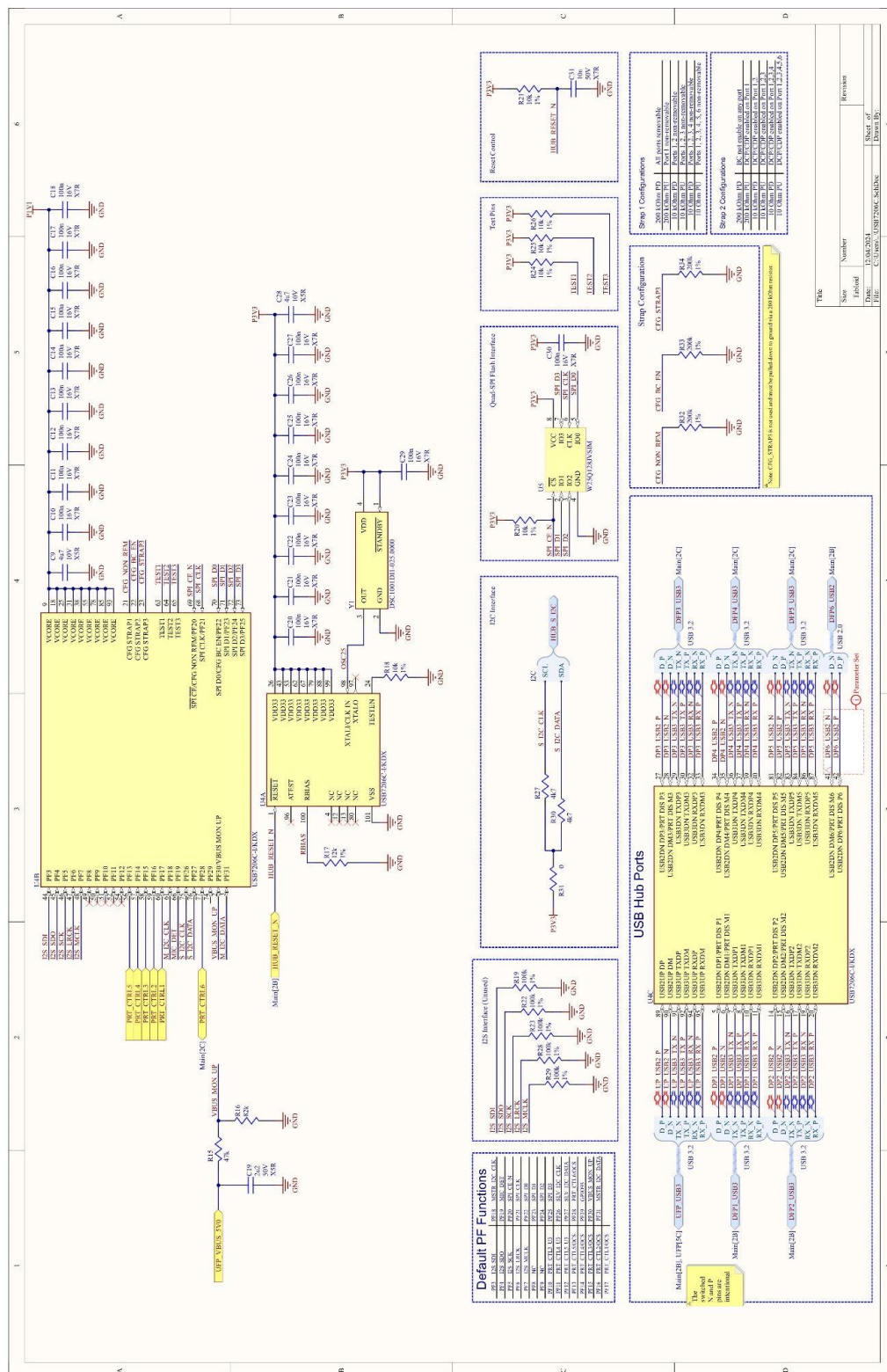


Figure A5. DFP Schematic

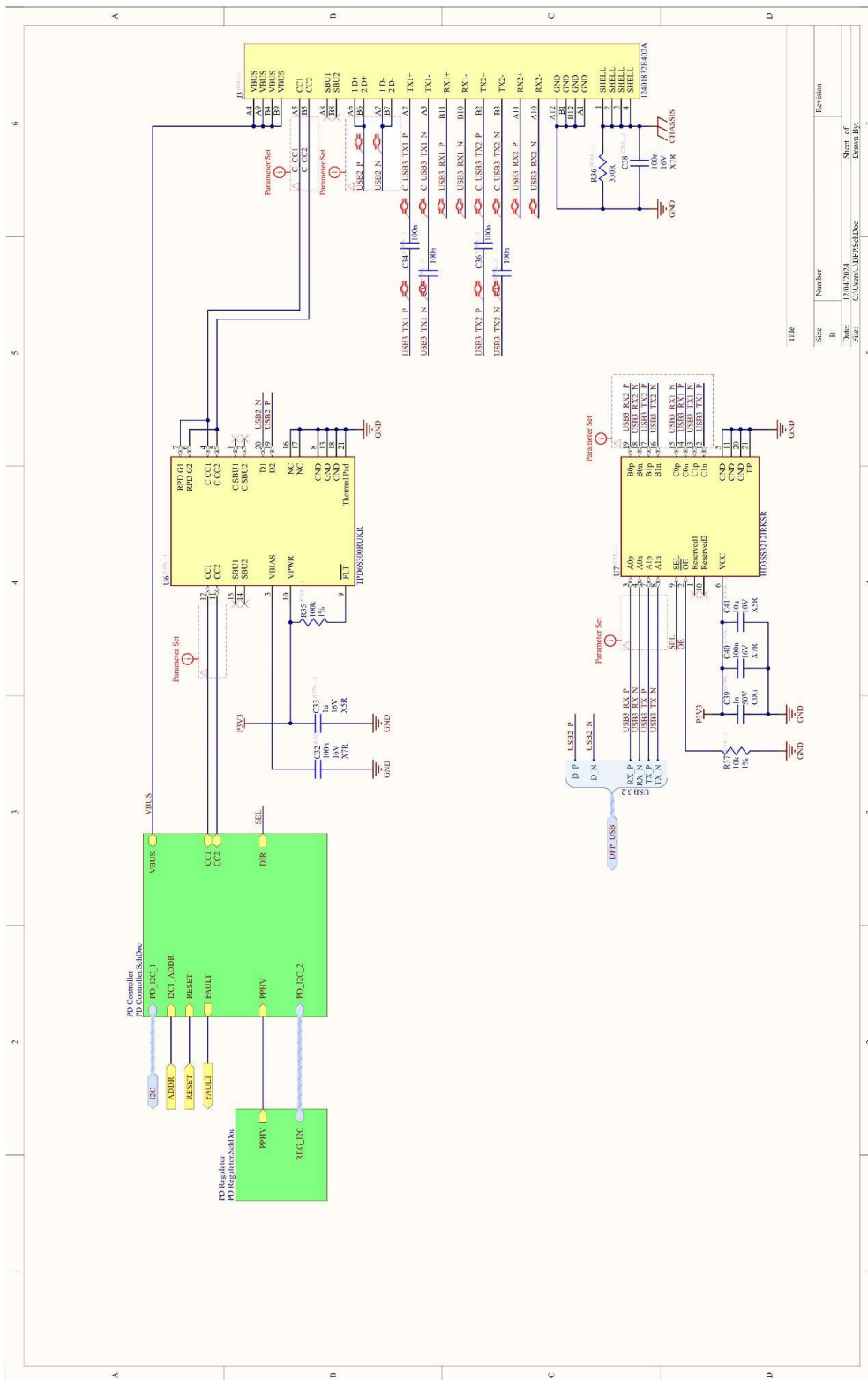


Figure A6. PD Controller Schematic

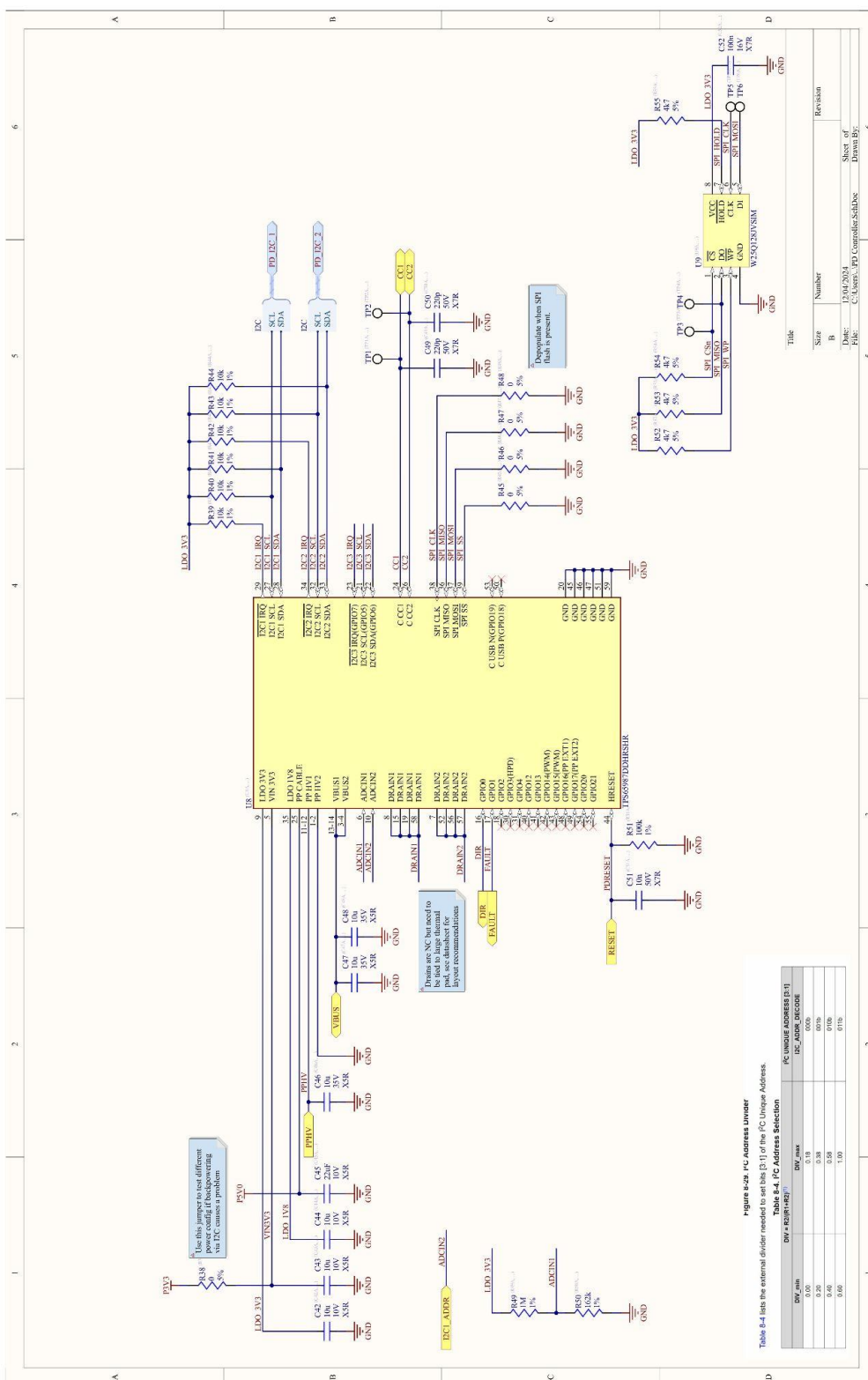


Figure A7. PD Regulator Schematic

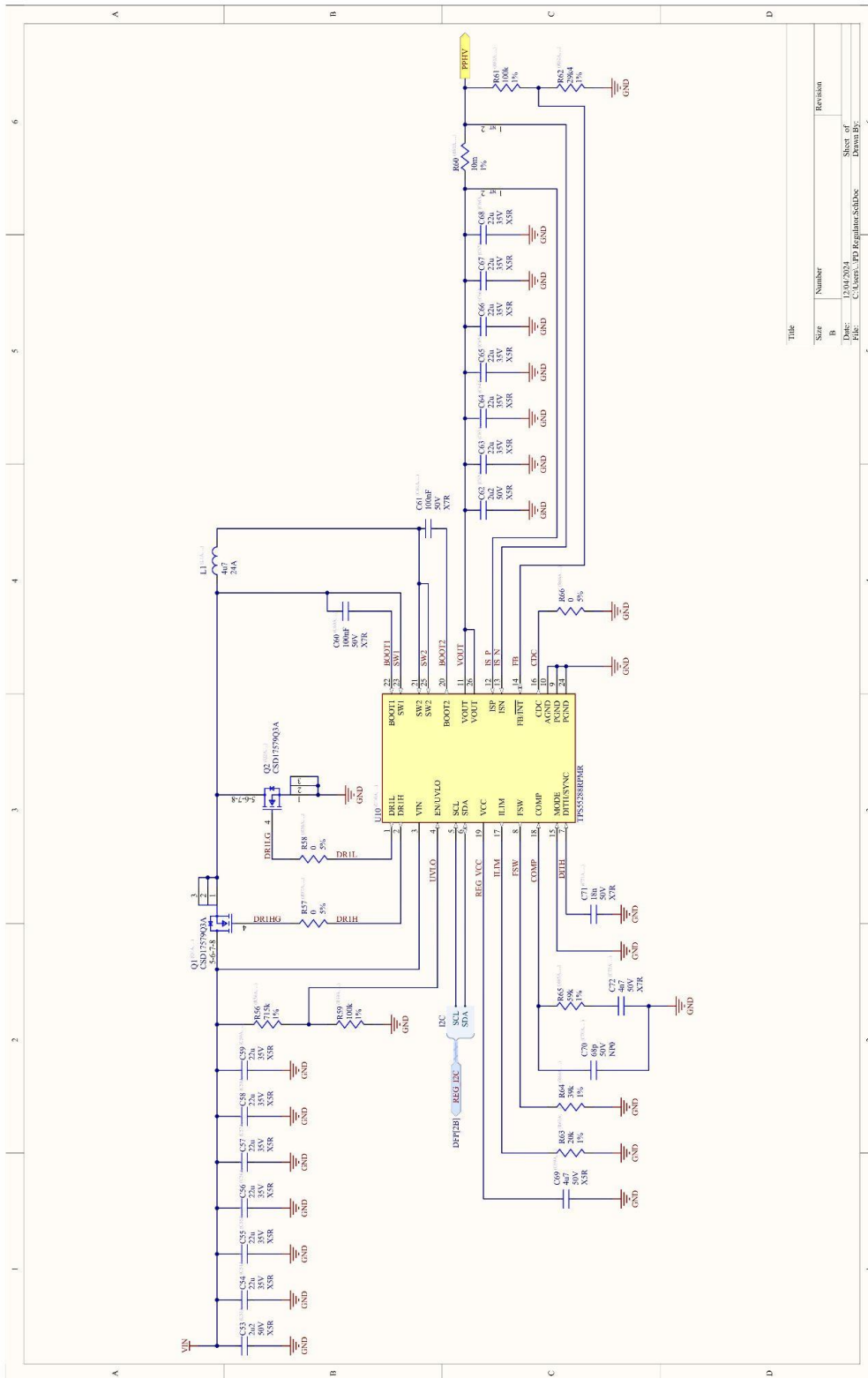
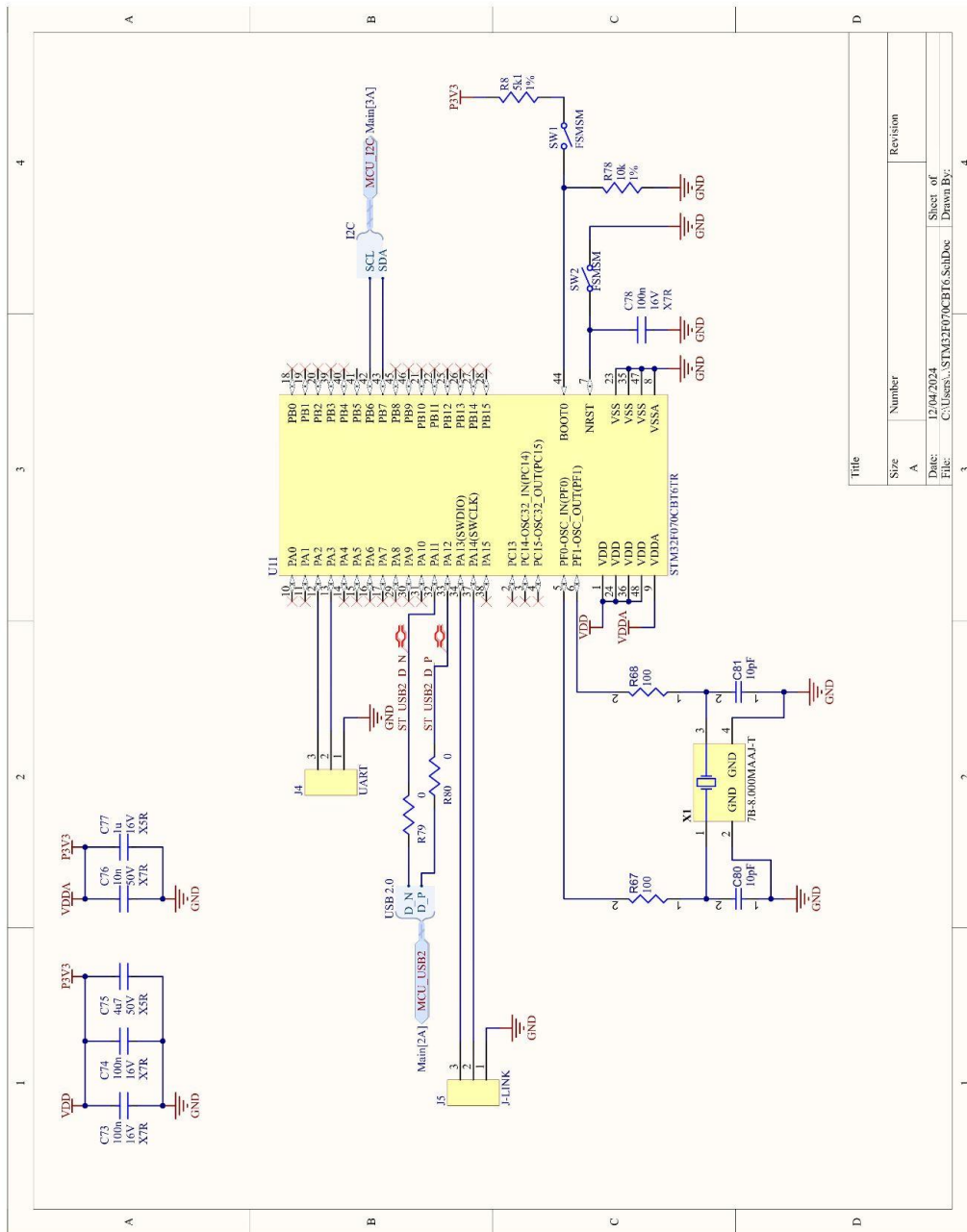


Figure A8. _____ Schematic



Appendix B: JLC Manufacturing Capabilities

Drilling	
Drill Diameter	0.15 – 6.3 mm
Hole size Tolerance (Plated)	Through-holes: +0.13 / -0.08 mm Press-fit holes : ± 0.05 mm
Hole size Tolerance (Non-Plated)	± 0.2 mm
Min. Via hole size/diameter	0.15mm / 0.25mm
Min. Non-plated holes	0.50mm
Min. Plated Slots	0.5mm
Min. Non-Plated Slots	1.0mm
Via Hole-to-Hole Spacing	0.2mm
Pad Hole-to-Hole Spacing	0.45mm
Traces	
Min. track width and spacing (1 oz)	0.10 / 0.10 mm
Min. track width and spacing (2 oz)	0.16 / 0.16 mm
Track width tolerance	$\pm 20\%$
PTH annular ring	≥ 0.20 mm
NPTH pad annular ring	≥ 0.45 mm
Same-net track spacing	0.25mm
Inner layer via hole to copper clearance	0.2mm
Inner layer PTH pad hole to copper clearance	0.3mm
Pad to track clearance	0.1mm
SMD pad to pad clearance (different nets)	0.15mm
Via hole to Track	0.2mm
PTH to Track	0.28mm min / 0.35mm rec
NPTH to Track	0.2mm
Soldermask	
Soldermask Expansion	0.038mm
Soldermask bridge	0.10mm
Solder mask ink thickness	$\geq 10\mu\text{m}$

Outline	
Routed edges to trace clearance	0.2mm
Silkscreen	
Minimum Line Width	6 mil (0.153mm)
Minimum text height	40 mil (1.0mm)
Character width to height ratio	1:6
Hollow-carved Character width to height ratio	1:6
Pad To Silkscreen	0.15mm

Appendix C: Bill of Materials

A	B	C	D	E	F	G	H
Comment	Description	Designator	Footprint	LibRef	Quantity	LCSC Part #	
100nF 16V 0402	16V 100nF X7R ±10% 0402	C1, C2, C7, C8, C10, C11	CAPC1005X06N	CMP-001-00088-1, CMP-001-00107-1, CMP-001-00117-1	44	C1525	
100nF 10V 0201	10V 100nF X5R ±10% 0201	C3, C4, C5, C6, C34A, C35	CAPC0603X03N	CMP-001-00107-1, CMP-001-00117-1	20	C76934	
4.7uF 10V 0402	10V 4.7uF X5R ±10% 0402	C9, C28	CAPC1005X06N	CMP-001-00117-1	2	C368809	
2.2uF 50V 0805	50V 2.2uF X5R ±10% 0805	C19, C53A, C53B, C53C	CAPC2012X14N	CMP-001-00094-2, CMP-001-00116-1	9	C377773	
10n 50V X7R 0402	50V 10nF X7R ±10% 0402	C31, C51A, C51B, C51C	CAPC1005X06N	CMP-001-00116-1	5	C60133	
1uF 16V X5R 0402	16V 1uF X5R ±10% 0402	C33A, C33B, C33C, C33	CAPC1005X06N	CMP-001-00098-2, CMP-001-00103-2	5	C29266	
1nF C0G 0603	C0G 1nF 50V ±5% 0603	C39A, C39B, C39C, C39	CAPC1608X08N	CMP-001-00103-2	4	C342541	
10uF 10V 0402	10uF 10V 0402 20% X5R	C41A, C41B, C41C, C41	CAPC1005X06N	CMP-001-00084-1, CMP-001-00110-1	16	C527590	
22uF 10V 0603 X5R	10V 22uF X5R ±20% 0603	C45A, C45B, C45C, C45	CAPC1608X08N	CMP-001-00110-1	4	C86295	
10uF 35V 0603	35V 10uF X5R ±20% 0603	C46A, C46B, C46C, C46	CAPC1608X08N	CMP-001-00102-2	18	C194427	
220pF 50V 0402	50V 220pF X7R ±10% 0402	C49A, C49B, C49C, C49	CAPC1005X06N	CMP-001-00115-1	8	C107025	
22uF 35V 1210 X5R	35V 22uF X5R ±20% 1210	C54A, C54B, C54C, C54	CAPC3225X18N	CMP-001-00092-1, CMP-001-00093-1	54	C778721	
100nF 50V X7R 0603	50V 100nF X7R ±10% 0603	C60A, C60B, C60C, C60	CAPC1608X08N	CMP-001-00093-1	8	C1591	
4.7uF 50V 0805	50V 4.7uF X5R ±10% 0805	C69A, C69B, C69C, C69	CAPC2012X14N	CMP-001-00089-1	5	C98192	
68pF 50V 0402	50V 68pF NPO ±1% 0402	C70A, C70B, C70C, C70	CAPC1005X06N	CMP-001-00113-1	4	C389103	
18nF 50V 0402	50V 18nF X7R ±10% 0402	C71A, C71B, C71C, C71	CAPC1005X06N	CMP-001-00111-2	4	C576740	
4.7nF 50V X7R 0402	50V 4.7nF X7R ±10% 0402	C72A, C72B, C72C, C72	CAPC1005X06N	CMP-001-00114-1	4	C1538	
10n 50V X7R 0603	50V 10nF X7R ±10% 0603	C76	CAPC1608X08N	CMP-001-00106-1	1	C100042	
Capacitor 10 pF +/- 5%	Chip Capacitor, 10 pF, C80, C81		CAPC1005X5X25ML05	CMP-001-00013-6	2		
SMAJ16CA	26V 17.8V Bidirectional	D1	DO-214AC	CMP-004-00043-1	1		
3557-2	Fuse Block 30 A 500V 1F1	F1	3557-2-Footprint-1	CMP-00030-00001-1	1	C19727304	
1812L160/12DR	POLY-FUSE(R) Resettable	F2	LFUS-1812L16012_V	CMP-005-00006-3	1		
12401832E402A	Conn USB Type C RCP 2J1, J3A, J3B, J3C, J3D		FP-12401832E402A-MFG	CMP-002-00130-1	5	C464604	
USB 3 Type A	USB Connector, 9 Contact	J2	692121030100	CMP-002-00126-1	1		
3x0.1" Header	3x0.1" pitch header	J4, J5	3x0.1" Header-Footprint	CMP-002-00078-3	2		
172448-0008	Rectangular Power Connector	J6	FP-172448-0008-MFG	CMP-002-00125-2	1		
4.7uH 24A 20%	20A 4.7uH Integrated Inductor	L1A, L1B, L1C, L1D	INDP134126X58N	CMP-006-00022-1	4	C526021	
10uH 2.5A SL0420-100H	2A 10uH ±20% 2.5A SMD	L2, L3, L4	10uH 2.5A SL0420-100H	CMP-006-00023-1	3		
CS017579Q3A	MOSFET N-CH 30V 35A	Q1A, Q1B, Q1C, Q1D, Q1E	FP-DNH0008A-MFG	CMP-013-00041-1	8	C97376	
200k 1% 0402	62.5mW Thick Film Resistor	R1, R32, R33, R34	RESC1005X04M	CMP-009-00220-2, CMP-009-00225-1	4	C114763	
910k 1% 0402	62.5mW Thick Film Resistor	R2	RESC1005X04M	CMP-009-00225-1	1	C137927	
0 Ohm 1% 0603	62.5mW Thick Film Resistor	R3, R4, R5, R6, R31, R75	RESC1508X05M	CMP-009-00190-2	7	C60485	
330R 1% 0402	62.5mW Thick Film Resistor	R7, R14, R36A, R36B, R36C	RESC1005X04M	CMP-009-00201-3	6	C105875	
5k1 1% 0603	100mW Thick Film Resistor	R8	RESC1508X05M	CMP-009-00204-2	1	C105580	
22K2F	22K 0.063W 1% 0402 (1R)	R9	RESC0402(1005)_L	CMP-009-00081-3	1		
10k 1% 0402	62.5mW Thick Film Resistor	R10, R11, R12, R13, R18	RESC1005X04M	CMP-009-00187-1, CMP-009-00226-1	38	C60490	
47k 1% 0402	62.5mW Thick Film Resistor	R15	RESC1005X04M	CMP-009-00226-1	1	C163440	
82k 1% 0402	62.5mW Thick Film Resistor	R16	RESC1005X04M	CMP-009-00227-1	1	C137932	
12k 1% 0402	62.5mW Thick Film Resistor	R17	RESC1005X04M	CMP-009-00219-1	1	C114760	
100k 1% 0402	62.5mW Thick Film Resistor	R19, R22, R23, R28, R29	RESC1005X04M	CMP-009-00218-1, CMP-009-00197-1, CMP-009-00232-1	21	C60491	
4.7k 5% 0402	62.5mW Thick Film Resistor	R27, R30, R52A, R52B, R52C	RESC1005X04M	CMP-009-00197-1, CMP-009-00232-1	18	C105870	
0 Ohm 1% 0402	62.5mW Thick Film Resistor	R38A, R38B, R38C, R38	RESC1005X04M	CMP-009-00232-1	28	C60485	
1M 1% 0402	62.5mW Thick Film Resistor	R49A, R49B, R49C, R49	RESC1005X04M	CMP-009-00240-1	4	C138033	
162k 1% 0402	62.5mW Thick Film Resistor	R50A, R50B, R50C, R50	RESC1005X04M	CMP-009-00217-3	5	C54068	
715k 1% 0402	100mW Thick Film Resistor	R56A, R56B, R56C, R56	RESC1005X04M	CMP-009-00228-2	4	C2480300	
10m 1W 1% 1206	1W 10mΩ ±70ppm/°C	R60A, R60B, R60C, R60	R1206	CMP-009-00229-1	4	C105362	
29k4 1% 0402	62.5mW Thick Film Resistor	R62A, R62B, R62C, R62	RESC1005X04M	CMP-009-00230-2	4	C26973	
20k 0402 1%	62.5mW Thick Film Resistor	R63A, R63B, R63C, R63	RESC1005X04M	CMP-009-00185-3	4	C93942	
39k 1% 0402	62.5mW Thick Film Resistor	R64A, R64B, R64C, R64	RESC1005X04M	CMP-009-00231-1	4	C137995	
59k 1% 0402	62.5mW Thick Film Resistor	R65A, R65B, R65C, R65	RESC1005X04M	CMP-009-00233-1	4	C163459	
100R2F	100R 0.063W 1% 0402 (1R)	R67, R68	RESC0402(1005)_L	CMP-009-00065-2	2		
0 Ohm 1% 0805	125mW Thick Film Resistor	R69, R72, R75	RESC2013X06M	CMP-009-00237-1	3	C96345	
30.9k 1% 0402	62.5mW Thick Film Resistor	R71	RESC1005X04M	CMP-009-00214-2	1	C185428	
180k 1% 0402	62.5mW Thick Film Resistor	R73	RESC1005X04M	CMP-009-00238-1	1	C144794	
62k 1% 0402	62.5mW Thick Film Resistor	R74, R77	RESC1005X04M	CMP-009-00239-1	2	C137950	
24.9k 1% 0402	62.5mW Thick Film Resistor	R76	RESC1005X04M	CMP-009-00200-2	1	C25874	
FSMSM	Switch Tactile OFF (ON)	SW1, SW2	TECO-FSMSM-2_V	CMP-011-00013-1	2		
SMT Test Point 5015	PC Test Point, Miniature	TP7, TP8, TP9	New footprint 1	CMP-00030-00002-1	3	C5199798	
USBLC6-2SC6	Trans Voltage Suppressor	U1, U3	SOT95P280X145-6N	CMP-015-00019-2, CMP-001-00008-5	2	C7519	
HD3S3220IRNHT	10-Gbps USB 3.1 Type-C	U2	FP-RNH0030A-MFG	CMP-001-00008-5	1	C165155	
USB7206C-I/KDX	USB3.1 GEN2 Hub, no fan	U4	FP-C04-407-IPC_A	CMP-001-0012-2	1		
TPD6S300RUKR	WQFN-20-EP(3x3) ESD	U6A, U6B, U6C, U6D	FP-RUK0020B-MFG	CMP-019-00064-3	4	C2649810	
HD3S3212IRKSR	2-channel 10-Gbps 2:1	U7A, U7B, U7C, U7D	FP-RKS0020A-MFG	CMP-001-0010-5	4		
TPS65987DDHRSHR	USB Type-C™ and USB	U8A, U8B, U8C, U8D	FP-RSH0056E-MFG	CMP-019-00062-1	4		
TPS55288RPMR	36V, 16A buck-boost converter	U10A, U10B, U10C, U10	FP-RPM0026A-MFG	CMP-019-00063-2	4	C2864583	
STM32F070CBT6TR	Mainstream ARM Cortex-M0	U11	STM-LQFP48_N	CMP-020-00035-1	1		
AP63201WU-7	Switching Regulator	U12, U13, U14	FP-TSOT26-IPC_B	CMP-019-00068-1	3	C2071044	
7B-8.000MAAJ-T	SMD Seam Sealing Crystal	X1	TXC-7B-4_V	CMP-003-00013-2	1		
DSC1001D11-025.0000	MEMS OSC XO 25.0000MHz	Y1	FP-CDFN2520-4LD-PL	CMP-06255-000046-1	1		